AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM358
Machine Learning & Artificial intelligence

Midterm Examination (MTA)
Summer Semester 2024/2025

Date: 7/Aug./2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 120 minutes

Instructions:

- Total Marks: **60**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

AOU EXAM

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Part -I: Question 1: Choose the best correct answer of all the following: (15 Marks)

1 .....means "How long does it take to find a solution?"

A Completeness
B Time complexity
C Optimality
D Space complexity

2 ..... is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators

A An Agent
B A computer
C A machine
D None of the above

3 ..... is a measure of how specific that schema is.

A Order
B dimension
C defining length
D None of the above

4 ..... tend to provide more information, but at the expense of increased power consumption and with a danger of interference when multiple sensors are used at the same time.

A mobile sensors
B rigid sensors
C Active sensors
D Passive sensors

5 ..... finds shallowest solution because always works on shallowest nodes first

A BFS
B A*
C DFS
D Best first

6 Chess game is an example of .....games.

A random
B deterministic
C stochastic
D None of the above

7 ..... can apply their effectors further afield than anchored manipulators can, but their task is made harder because they don't have the rigidity that the anchor provides.

A Mobile robot
B Manipulators
C mobile manipulator
D None of the above

8 ..... knowledge deals with factoid questions

A approximative
B Procedural
C Instructional
D Declarative

9 Stone game is an example of .....games.

A random
B stochastic
C deterministic
D None of the above

10 .....can be defined as "is a tiny 'world' comprising an (infinitely large) flat table on which sit a set of children's building blocks. "

A smart world
B Block world
C Block Chain
D None of the above

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Part-II: Answer all of the following questions (45 Marks).

Question 2: (15 marks)

A tree can be defined by two vectors. The first vector is the parents vector -s. The second vector is the children vector -t. Each entry in t matrix is a child to the corresponding parent in the same index in the s matrix. Assuming s, t in order for a given tree as:

s	A	A	B	B	B	C	C	C	D	D	F	F	F	G	G	G	H	I
t	B	C	D	39	F	G	H	I	27	3	15	6	20	2	49	45	4	28

Given the aforementioned tree, use α-β pruning minimax algorithm to evaluate each node in the tree.

B. Explain how the mechanism of interaction contributes to the development of natural intelligence in species.

Question 3: (15 marks)

A tree can be defined by two vectors. The first vector is the parents vector -s. The second vector is the children vector -t. Each entry in t matrix is a child to the corresponding parent in the same index in the s matrix. Assuming s, t in order for a given tree as:

s	A	A	B	B	C	C	D	D	E	E	F	F	G	G	G
t	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P

Given the aforementioned tree, apply DFS to traverse the tree showing all agenda content and traced paths until reaching the goal state if exist assuming that the goal state is M

B. Discuss the concept of clobbering in partial-order planning systems and explain how it affects planning efficiency.

Question 4: (15 marks)

A

Having the following routing table at node 5

	N66	**N40**	**N58**	**N79**	**N100**
N93	0.18	0.25	0.24	0.11	0.34
N28	0.22	0.52	0.52	0.37	0.26
N2	0.16	0.05	0.10	0.10	0.25
N38	0.44	0.17	0.14	0.43	0.16

Having started at node 100 and reaching node 5 via node 38.Assuming it has been in the network for 1.894450 time units, what effect will it have on the routing table on node 5 assume a and b equals 0.010000 and 0.060000 in order for ABC model

B. Explain the concept of a constraint satisfaction problem (CSP) and describe the components of a CSP. Provide an example of a CSP.

End of questions

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